

SIN'GYE, Korea, Democratic People's Republic ofWMO: **470670**Lat: **38.500N**Lon: **126.533E**Elev: **318**StdP: **14.53**Time Zone: **9.00 (E09)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	8.2	-8.9	3.4	9.0	-4.9	4.3	12.8	19.0	23.3	16.7	21.2	2.3	320	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.2	88.8	76.1	86.2	74.9	84.0	73.7	79.4	85.8	78.0	83.5	76.7	81.1	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.6	145.4	83.0	76.4	139.3	81.2	75.3	134.3	79.6	43.3	86.2	41.8	83.8	40.5	81.4	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	14.9	12.3	-0.4	92.8	5.8	3.0	-4.7	94.9	-8.1	96.6	-11.4	98.3	-15.6	100.4		
(5)			-0.9	82.3	5.4	2.4	-4.8	84.0	-8.0	85.4	-11.1	86.7	-15.0	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	22.6	28.5	39.3	50.9	62.4	70.9	75.7	76.4	67.8	55.6	41.4	27.1
	DBStd	19.69	7.24	7.21	6.32	6.39	5.26	4.08	3.98	4.20	4.65	6.25	8.44	7.93
	HDD50	2864	851	602	338	64	0	0	0	0	0	23	277	709
	HDD65	5882	1316	1022	797	424	115	6	0	0	25	294	709	1174
	CDD50	3472	0	0	5	91	384	626	797	819	534	197	19	0
	CDD65	1014	0	0	0	1	33	181	332	354	109	4	0	0
	CDH74	6923	0	0	0	17	346	1141	2142	2703	546	28	0	0
	CDH80	1741	0	0	0	1	61	275	562	785	56	1	0	0
Wind	WSAvg	3.2	2.8	3.5	4.3	5.2	4.5	3.1	3.3	2.1	1.9	2.2	2.9	2.8
Precipitation	PrecAvg	49.6	0.6	0.9	1.0	2.3	4.0	5.0	15.9	11.6	4.6	1.9	1.7	0.7
	PrecMax	87.4	1.7	2.8	2.7	5.3	10.3	17.7	35.4	27.7	13.6	6.5	4.3	2.2
	PrecMin	20.0	0.0	0.0	0.1	0.2	0.8	1.4	5.0	1.5	0.3	0.2	0.2	0.1
	PrecStd	13.5	0.5	0.7	0.7	1.3	2.4	3.6	7.5	7.4	3.6	1.8	1.1	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	52.6	64.6	76.6	85.1	89.5	93.2	92.3	84.5	77.4	64.4	49.0
		MCWB	37.4	46.2	50.0	58.6	64.9	69.6	78.2	80.3	71.8	63.7	56.8	44.2
	2%	DB	38.7	46.7	58.8	71.3	81.3	86.1	88.1	89.3	81.7	73.3	60.0	45.0
		MCWB	33.8	40.3	47.9	55.3	63.2	69.3	76.9	78.2	70.0	61.5	53.6	41.6
	5%	DB	35.9	43.0	54.2	67.3	77.8	83.4	85.3	86.8	79.5	70.1	56.7	41.7
		MCWB	32.0	37.2	45.6	53.3	61.9	68.7	75.9	77.5	68.5	59.6	51.1	38.6
	10%	DB	33.7	39.6	50.4	63.2	74.3	80.8	82.9	84.4	77.3	67.1	53.8	38.5
		MCWB	30.5	34.9	42.8	51.3	60.9	68.0	74.9	76.1	67.4	57.9	49.2	35.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	48.3	53.3	62.4	69.1	75.4	81.4	82.4	75.4	66.5	59.5	46.0
		MCDB	41.2	50.0	58.5	70.0	79.7	82.6	88.4	88.8	79.6	73.2	62.5	47.7
	2%	WB	35.1	41.4	49.4	57.6	66.4	73.0	79.1	80.0	73.3	63.6	54.8	42.6
		MCDB	37.5	45.7	56.1	66.6	75.9	80.4	85.4	86.6	77.5	70.2	58.4	44.3
	5%	WB	32.8	38.0	46.4	55.1	64.4	71.5	77.6	78.7	71.6	61.5	52.2	38.8
		MCDB	35.2	41.7	53.4	64.2	73.2	78.4	82.9	84.8	77.0	67.7	56.1	41.3
	10%	WB	30.6	35.3	43.8	52.9	62.6	70.1	76.4	77.5	69.7	59.2	49.7	35.6
		MCDB	33.4	39.3	49.5	60.7	71.5	76.8	80.7	82.4	75.2	65.6	52.9	38.0
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.1	17.6	20.0	19.9	16.4	11.5	13.2	17.4	20.4	15.9	14.9
		MCDBR	18.1	20.0	23.2	26.4	26.1	22.4	16.6	16.4	19.9	23.8	19.3	16.7
		MCWBR	14.2	14.3	14.0	13.2	11.5	9.4	7.8	7.7	9.8	12.8	13.2	13.4
	5% WB	MCDBR	15.5	17.1	20.2	22.0	20.9	16.9	12.6	13.8	15.0	20.0	16.7	15.3
		MCWBR	13.2	13.3	13.3	13.0	11.2	8.9	7.1	7.9	9.6	12.5	12.8	13.2
Clear-Sky Solar Irradiance	taub		0.415	0.484	0.556	0.619	0.627	0.705	0.676	0.575	0.504	0.511	0.483	0.421
	taud		1.928	1.814	1.707	1.627	1.651	1.563	1.676	1.900	2.005	1.872	1.877	1.952
	Ebn at Noon		224	224	222	218	220	203	208	227	235	213	199	210
	Edh at Noon		44	56	69	80	79	86	77	60	51	52	46	40
All-Sky Solar Radiation	RadAvg		793	1016	1328	1547	1741	1629	1251	1357	1331	1115	751	685
	RadStd		46	73	117	117	176	166	177	179	134	69	76	51